Thoughts And Rambles On Probability And Statistics

Problem Set 7: The Law Of Large Numbers

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These are the end of chapter problems for Chapter 7 of the notes. Equation numbers refer to the full notes.

1. Let $X_1, X_2, \ldots$ be random variables with $\mathbb{E}[X_k] = \mu$, $\text{Var}[X_k] \leq C$ for all k, and $\text{Cov}(X_j, X_k) = 0$ for $j \neq k$. Show that $\bar{X}_n \to \mu$ in probability. Where exactly did the two line derivation use independence?

2. Let $U_1, U_2, \ldots$ be i.i.d. uniform on $[0, 1]$. Find the almost sure limit of the geometric mean $(U_1 U_2 \cdots U_n)^{1/n}$. Compare it with $\mathbb{E}[U_1]$.

3. (Computational) Use Monte Carlo to estimate $\int_0^1 e^{-x^2} dx$ with $n = 10, 10^2, \ldots, 10^6$ samples. Plot the absolute error against n on a log-log plot and compare with $\sigma/\sqrt{n}$, where $\sigma^2 = \text{Var}[e^{-U^2}]$.

4. Show that if $\mathbb{E}|X_1| < \infty$, then $\bar{X}_n \to \mu$ in L_1 as well. (Hint: show that $\{\bar{X}_n\}$ is uniformly integrable, and use chapter 6.)

5. (Kelly betting) You can bet any fraction $f \in [0, 1]$ of your wealth on a coin that lands heads with probability $p > 1/2$. If it lands heads you win the amount you bet, otherwise you lose it. Show that your wealth grows like $e^{ng(f)}$ almost surely, find $g(f)$, and show that the optimal fraction is $f^* = 2p - 1$. What happens if you bet everything every time?

6. Let $X_1, X_2, \ldots$ be independent with $\mathbb{P}(X_n = n^\alpha) = \mathbb{P}(X_n = -n^\alpha) = 1/2$. For which α does $\sum_n X_n/n$ converge almost surely? For which α does $\bar{X}_n \to 0$ almost surely?

7. Let X be symmetric with $\mathbb{P}(|X| > x) = \frac{e}{x \log x}$ for $x \geq e$. Show that $\mathbb{E}|X| = \infty$ but $x\,\mathbb{P}(|X| > x) \to 0$. Conclude that i.i.d. copies of X satisfy the weak law with $\mu = 0$, but not the strong law.

8. Let $X_1, X_2, \ldots$ be i.i.d. with $\mathbb{E}[X_1^2] < \infty$. Show that the sample variance $\frac{1}{n}\sum_{k=1}^n (X_k - \bar{X}_n)^2$ converges almost surely to $\text{Var}[X_1]$.